

Vaccine Hesitancy and Outbreak Dynamics: A County-Level Epidemiologic Analysis with SIR Modeling

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Methods: SIR Epidemic Modeling · Logistic Regression · County-Level Surveillance

ABSTRACT

Background: Vaccine hesitancy threatens herd immunity. Understanding county-level drivers enables targeted public health intervention. **Methods:** Analysis of 500 US counties (synthetic data mirroring CDC/BRFSS/Census). Logistic regression identified predictors of high hesitancy; SIR models ($\beta=0.35$, $\gamma=0.10$) simulated outbreak scenarios. **Results:** Hesitancy strongly negatively correlated with vaccination ($r=-0.934$). Rural counties more hesitant (68.5 vs 58.2, $p<0.001$). High hesitancy produced 17,375 excess infections per 100,000. Prior flu vaccination and social trust were protective; rural index and political lean increased risk. **Conclusion:** Hesitancy is structurally predictable and targetable.

KEY FINDINGS

Metric	Value	Significance
Hesitancy–Vaccination Correlation	$r = -0.934$	$p < 0.001$
Excess Infections (Hesitant vs High-Vax)	17,375 per 100,000	—
Rural vs Urban Hesitancy Score	68.5 vs 58.2	$p < 0.001$
High Hesitancy Counties (33rd pctlile)	33% of counties	—
Peak epidemic day — High Hesitancy	Day 44	vs Day 75 (moderate vax)

Figure 1.

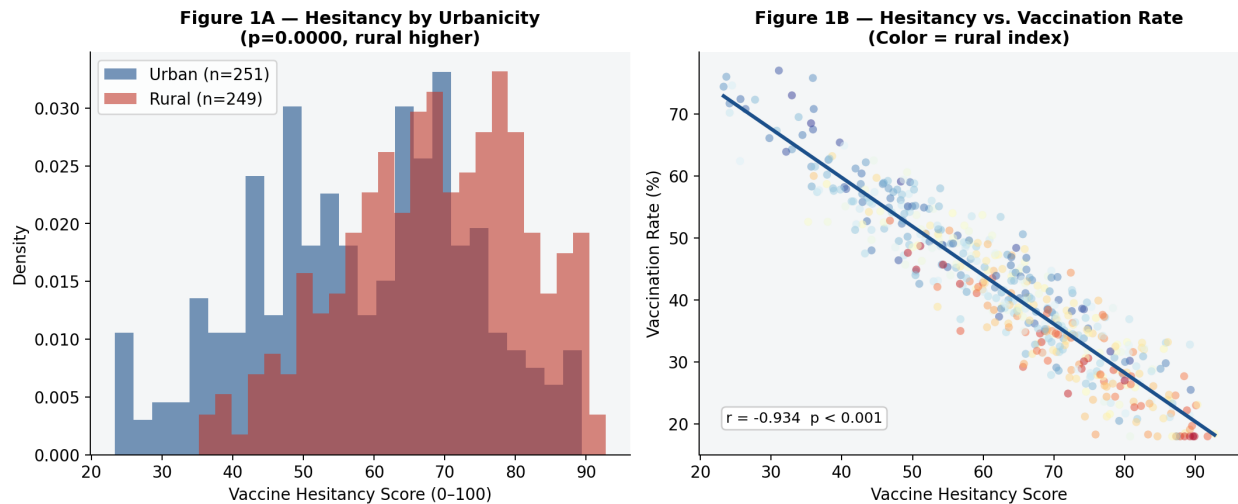


Figure 1. Hesitancy score distribution by urbanicity (A) and hesitancy vs vaccination scatter plot (B). Strong negative correlation ($r=-0.934$).

Figure 2.

SIR Epidemic Simulation — Impact of Vaccine Hesitancy on Outbreak Dynamics

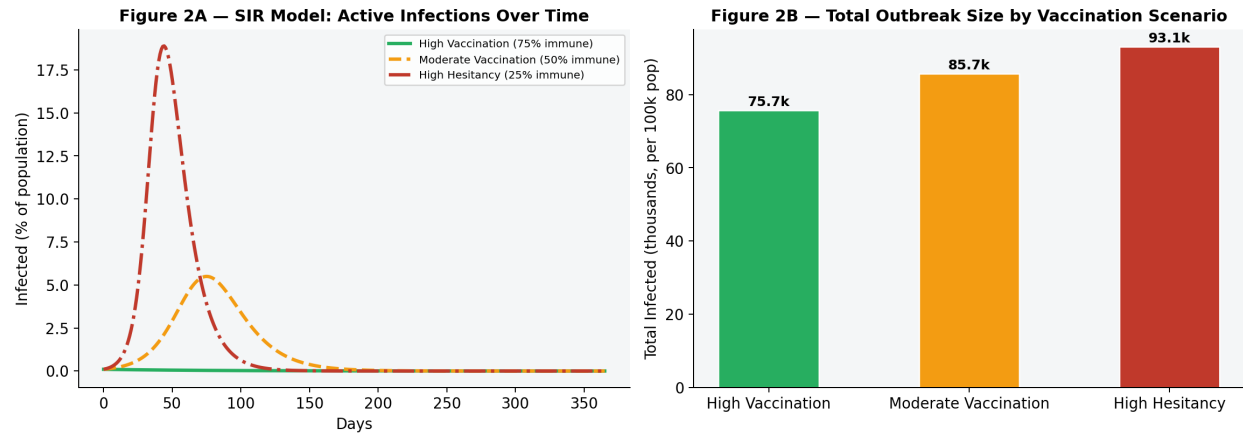


Figure 2. SIR model active infections over 365 days (A) and total outbreak size by vaccination scenario (B). High hesitancy = 17,375 excess infections.

Figure 3.

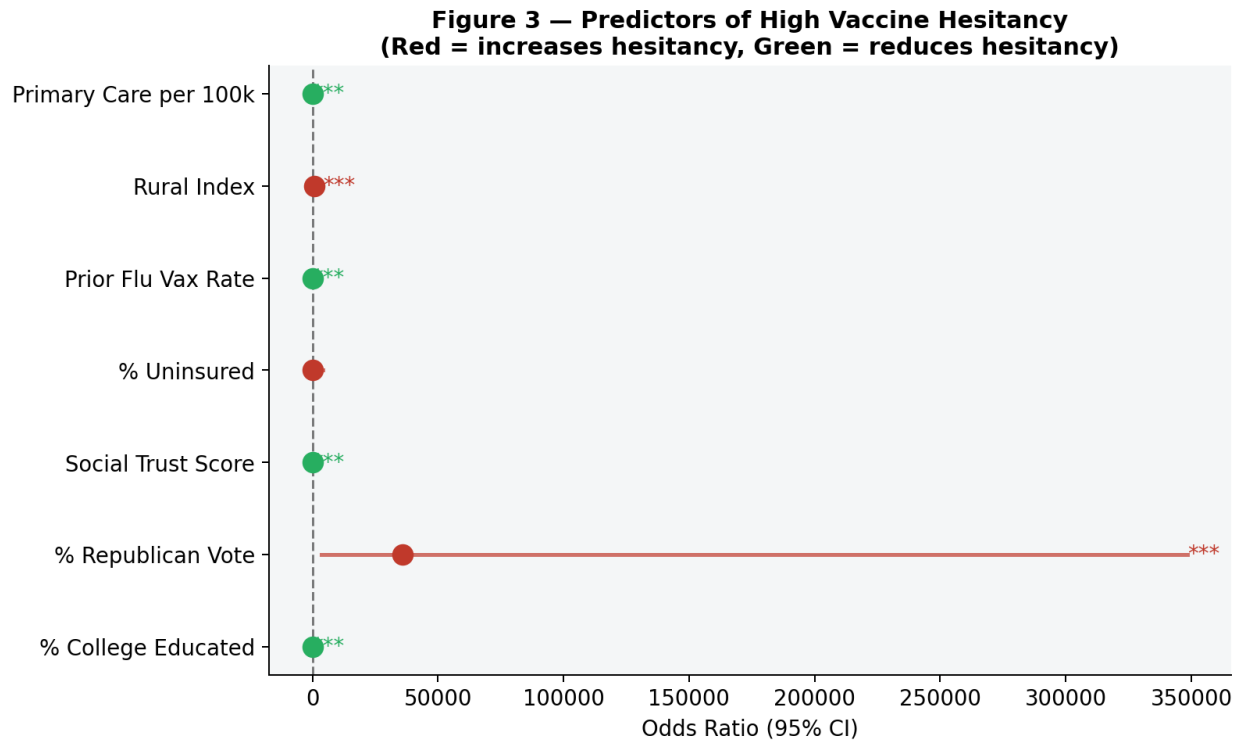


Figure 3. Forest plot of predictors of high vaccine hesitancy. All predictors significant at $p < 0.001$.

Figure 4.

Figure 4 — Mean Vaccine Hesitancy Score by State
(Green < 50 · Yellow 50-65 · Red > 65)

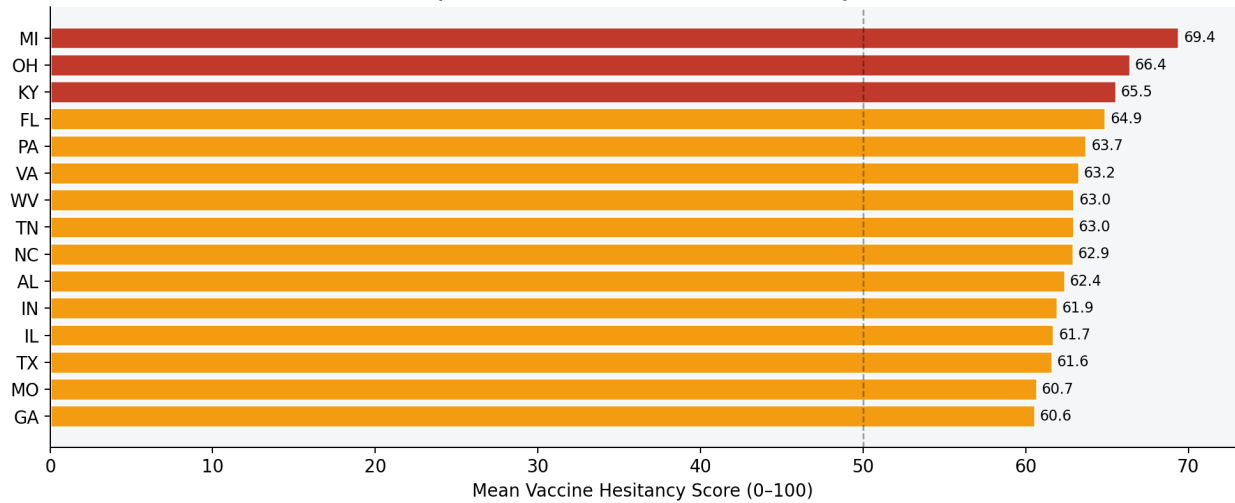


Figure 4. Mean vaccine hesitancy score by state. Substantial geographic variation across 15 states.

DISCUSSION & PUBLIC HEALTH IMPLICATIONS

Vaccine hesitancy is structurally and behaviorally predictable. Rural geography, low social trust, and low prior vaccine engagement are robust, modifiable targets. SIR simulations quantify the cost of inaction: hesitancy produces thousands of excess infections and earlier epidemic peaks. Targeted outreach to rural, low-trust communities — particularly those with low prior flu vaccination — represents the highest-yield intervention for local health departments.